

WOLF POINT 29 ENE, MT, USA

WMO: 742320

Lat: 48.3082N

Lon: 105.1018W

Elev: 636

StdP: 93.92

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 94060

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-27.5	-33.7	0.2	-30.6	-30.4	0.2	-27.2	11.0	-1.5	10.0	-3.0	1.5	N/A	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	34.5	18.1	32.4	18.2	30.3	17.6	21.0	28.4	19.7	27.6	18.8	27.1	3.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	14.4	25.1	17.0	13.1	23.1	15.8	12.1	21.8	63.8	28.2	59.0	27.7	55.6	26.7	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	7.8	6.8	-36.3	38.1	4.3	2.4	-39.4	39.8	-41.9	41.2	-44.4	42.5	-47.5	44.2
			-37.1	23.5	3.0	2.0	-39.3	24.9	-41.0	26.1	-42.7	27.2	-44.9	28.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-10.4	-10.1	-2.0	5.5	11.3	16.7	21.0	19.5	13.7	5.6	-2.6	-9.7
	DBStd	12.77	8.61	8.16	7.79	5.36	4.76	3.65	3.39	3.79	4.86	5.48	6.83	7.86
	HDD10.0	2927	633	564	373	151	42	2	0	0	18	157	377	611
	HDD18.3	5087	892	797	629	386	223	73	12	32	152	396	627	869
	CDD10.0	1086	0	0	2	16	81	201	339	296	131	20	1	0
	CDD18.3	203	0	0	0	0	4	22	93	69	14	1	0	0
	CDH23.3	3561	0	0	1	19	136	442	1428	1140	366	29	0	0
	CDH26.7	1506	0	0	0	3	30	136	661	523	149	5	0	0
Wind	WSAvg	2.8	2.7	2.6	3.0	3.3	3.3	2.9	2.6	2.6	2.6	2.8	2.5	2.5
Precipitation	PrecAvg	349	11	7	13	25	54	76	55	41	24	22	11	10
	PrecMax	517	30	18	41	55	150	155	203	158	64	78	43	22
	PrecMin	222	3	2	2	9	16	24	16	1	6	3	1	1
	PrecStd	61	7	4	8	13	35	30	36	32	17	18	9	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	10.8	21.2	26.2	29.9	33.4	37.7	37.5	34.3	26.9	17.7	9.2
		MCWB	4.2	5.0	10.1	13.2	15.7	18.9	17.9	18.1	17.6	14.7	9.6	3.8
	2%	DB	5.7	6.2	16.3	21.7	26.8	30.2	34.9	34.3	30.8	22.5	13.3	5.3
		MCWB	2.6	2.5	8.1	10.9	14.7	18.0	18.5	17.6	15.8	13.0	7.2	2.0
	5%	DB	3.5	3.3	12.4	18.9	24.5	27.9	32.7	32.0	27.8	18.6	10.2	3.2
		MCWB	1.2	1.1	6.1	9.1	13.7	17.0	18.8	17.5	15.0	10.2	5.3	0.8
	10%	DB	1.5	1.1	8.9	16.2	21.6	25.8	30.6	29.6	24.4	15.4	7.1	1.1
		MCWB	-0.2	-0.3	4.5	7.7	12.3	16.0	18.6	17.1	13.9	8.9	3.4	-0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	5.1	10.2	13.3	18.0	22.6	23.8	21.1	20.2	15.3	10.0	4.4
		MCDB	8.6	9.8	19.7	24.4	25.4	28.9	31.1	29.6	28.9	24.7	17.5	8.5
	2%	WB	2.7	2.7	8.6	10.7	16.2	19.5	21.7	19.9	17.6	13.3	7.6	2.2
		MCDB	5.6	5.5	16.2	19.8	23.2	27.0	29.3	28.4	27.4	19.9	12.8	5.1
	5%	WB	1.3	1.3	6.7	9.1	14.8	18.0	20.3	18.9	15.9	11.1	5.2	0.9
		MCDB	3.4	3.3	12.7	16.8	22.3	25.6	27.7	28.0	24.4	17.3	9.3	2.9
	10%	WB	0.0	0.0	4.7	7.9	13.4	16.9	19.2	18.0	14.7	9.5	3.3	-0.6
		MCDB	1.7	1.6	9.1	14.8	19.8	24.1	26.9	27.2	23.1	15.2	6.1	1.0
Mean Daily Temperature Range	5% DB	MDBR	12.0	12.1	12.6	14.8	14.9	14.6	16.5	16.9	16.9	14.6	13.1	11.9
		MCDBR	13.5	14.6	18.3	20.6	19.5	18.6	20.0	20.8	22.6	20.6	17.8	14.3
		MCWBR	9.7	10.3	10.8	11.2	9.4	8.2	7.2	7.9	9.8	10.7	11.7	10.5
	5% WB	MCDBR	12.2	12.9	17.9	19.2	16.4	15.8	16.1	17.2	18.7	18.8	17.5	13.3
		MCWBR	9.7	10.1	10.7	10.6	8.3	7.9	7.3	6.9	8.5	10.3	11.8	10.6
Clear-Sky Solar Irradiance	taub	0.250	0.269	0.275	0.319	0.338	0.351	0.359	0.367	0.323	0.287	0.259	0.243	
	taud	2.472	2.463	2.543	2.425	2.424	2.430	2.425	2.377	2.469	2.560	2.535	2.502	
	Ebn at Noon	852	908	957	934	923	909	896	873	886	876	836	820	
	Edh at Noon	61	77	84	106	110	110	110	110	91	70	57	52	
All-Sky Solar Radiation	RadAvg	1.47	2.49	3.74	4.94	5.88	6.42	6.89	5.64	4.14	2.62	1.58	1.11	
	RadStd	0.13	0.17	0.24	0.23	0.47	0.38	0.32	0.38	0.37	0.26	0.18	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page